King Abdulaziz University

Faculty of Computing & Information Technology

Computer Science Department

Course: CPCS204 (Data Structures)

Term: 2nd Semester (Spring 2024-02)

Answer key (Version A)

Mid Exam - Wednesday 16 Oct, 2024 (1:00 - 2:20 pm)

Time: 90 minutes

Student ID: ____ **Section:** _______________

Student Name: _____________________________________

Course Coordinators: Dr. Samar Alkhuraiji, Dr. Ali Alkhathlan

Question 1 : _________ /33 +

Question 2 : _________ /25 +

Question 3 : _________ /30 +

Question 4 : _________ /12

Final Score: _______________ / 100

Question 1, Mixed Topics: _________ / 33 points

Part A: 10 Multiple Choice Questions (__________ /20) **[2 points each]**

1) What is the time complexity to access an element in an array, given its index?

a) O (1)
b) O (n)
c) O (log n)
d) O (n^2)

2) What is the primary disadvantage of arrays?

a) They can store multiple data types
b) Accessing elements takes O(n) time
c) The size of an array is fixed once declared
d) Inserting elements at the end is inefficient

3) Which of the following statements accurately compares Binary Search and Linear Search?

a) Binary Search can only be applied to sorted arrays, while Linear Search can be applied to both sorted and unsorted arrays
b) Linear Search is generally faster than Binary Search for large sorted arrays
c) Binary Search requires more memory than Linear Search due to its algorithmic complexity
d) Linear Search can find an element in O (log n) time, whereas Binary Search takes O (n) time for unsorted arrays

4) Which of the following data structures can be used to implement a Stack data structure?

a) Array
b) Singly linked list
c) Doubly linked list
d) All of the above

5) Which operation on a Singly linked list has the worst-case time complexity of O(n), with the first node as `head`?

a) Insert at the beginning
b) Insert at the end
c) Delete the first node
d) They are all the same

6) What is recursion in programming?

a) A function that calls itself
b) A function that calls other functions
c) A loop that repeats until a condition is met
d) A method for optimizing code

7) What happens if a recursive function does not have a proper base case?

a) It will run only once
b) It will compile but not execute
c) It will result in an infinite recursion, eventually causing a Stack overflow
d) It will return null or 0 by default

8) Which of the following is true about time complexity O (1)?

a) The execution time grows linearly with input size
b) The execution time remains constant regardless of input size
c) The execution time doubles as the input size doubles
d) The execution time grows logarithmically with input size

9) What is the time complexity of the push and pop operations in a Stack implemented using an a linked list?

a) O (n)
b) O (log n)
c) O (1)
d) O (n^2)

10) In a Stack implemented using an array, what is the initial value of the TOP pointer in an empty Stack?

a) 0
b) 1
c) -1
d) n

Part B: 5 Short Answers Questions (__________ /13)

1) **What will be printed given the following code segment for a given Singly linked list with the first node as `head`, `helpPtr = head`, and contains the following data items: 5 → 10 → 15**

[3 points]

```
void Enjoy1(LLnode helpPtr) {
    if (helpPtr == null)
        return;
    System.out.print(helpPtr.data + " ");
    Enjoy1(helpPtr.next);
    System.out.print(helpPtr.data + " ");
}
```

Output: ___**5 10 15 15 10 5** ____ 0.5 each

2) **What will be the content of a given Singly linked list with the first node as `head`, last node as `last` and containing the following data items: 10 → 20 → 30 after executing the function `Enjoy2(5)`.**

[3.5 points]

```
void Enjoy2(int value) {
    head = new LLnode(head.next.data, head);
    last.next = new LLnode(last.data, null);
    last = last.next;
}
```

Singly linked list content: **20 → 10 → 20 → 30→ 30** 1 + 0.5 + 0.5 + 0.5 + 1

3) **What will be the return value after executing the function `Enjoy3(3)`?** **[2.5 points]**

```
public static int Enjoy3(int n) {
    if (n == 1) return 1;
    else return 2 * Enjoy3(n - 1);
}
```

Return value:_________ **4**

4) **What will be printed given the following code segment?** **[2 points]**

```
int[] arr= {0, 1, 2, 3, 4, 5, 6, 7, 8, 9};
System.out.println(arr[arr[6]/2]);
```

Output: __________**3**

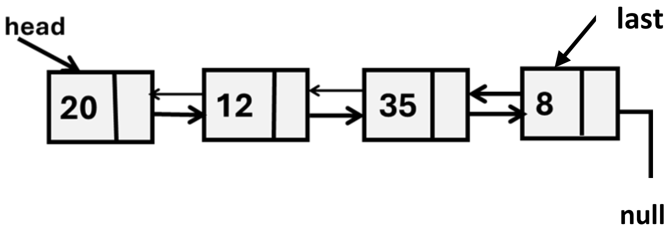
5) **Given a sorted array** `float[] Hights= {155.2, 156.1, 177.6, 188.2, 190.2, 195.3, 199.0, 200.1}`. **Using the Binary Search Algorithm, what is the number of comparisons required to find a `key value = 188.2` in the array?** **[2 points]**

Number of comparisons: _________**1**

Question 2, Linked List & Stack Topics: _________ / 25 points

Part A: Linked List Topic (__________ /16)

1) Consider the following sorted Doubly linked list, where `head`, and `last` are pointing to the first and last nodes in the list, respectively as follows: [________ /11 points]



Then answer the following questions.

a) What will be the return value after executing the function `Fun()`? [2 points]

```
int Fun() {
    int  T = 0;
    DLLnode current = head;
    while (current.next != null) {
        T++;
        current = current.next;
    }
    return T;
}
```

Return value: ________3

b) What will be printed as the result of executing the following code statement? ___________12

```
System.out.println(head.next.data);
```

[2 points]

c) What will be printed as the result of executing the following code statement? ___________12

```
System.out.println(head.next.next.prev.data);
```

[2 points]

d) What will be printed as the result of executing the following code statements? __________35

```
current = head.next;
System.out.println(current.next.next.prev.data);
```

[2 points]

e) What will be the content of the Doubly linked list above after executing the following code statements? [3 points]

```
last = last.prev;
last.next =  null;
```

Content of the Doubly linked list: ________________20 → 12→35

2) Complete the following boolean function `hasCycle(LLNode head)`. The `hasCycle` function should take the `head` of a Singly linked list as input and return `true` if the list contains a cycle, or `false` otherwise. A linked list has a cycle when a last node in the linked list points to the first node (`head` node), creating a loop.

Note: linked list with one node has no cycle **[________ /5 points]**

```
public  static boolean hasCycle(LLNode head) {
 if ( head == null)                    2 points
         return false;
  LLNode ptr = head;
  while (ptr != null) {
     if (ptr.next == head)             1.5 points
          return true;
     ptr = ptr.next;                   1.5 points
 }
 return false;
}
```

Part B: Stack Topic (__________ /9)

1) Evaluate the following postfix expression using a Stack. Additionally, you must show the contents of the Stack at the indicated points (A, B, and C) in the given postfix expression and give the final evaluation result. **[9 points]**

Postfix expression: 5 2 3 + 4 * * 6 10 - 3 + /

A (after +), B (after 6), C (after 3)

A	B	C
		3
5	6	-4
5	100	100

2 points A, B , 3 points C

Final Evaluation result: _____________-100

2 points final result

Question 3, Recursion Topic: _________ / 30 points

1) Complete the following recursive function `SumOfEven(int[] arr, int index)` that takes an array of integers and an index as parameters. The function should calculate and return the sum of all even numbers in the array starting from the given index to the end of the array. If the index is beyond the length of the array, the function should return 0.
Example: Given the array: arr = {1, 2, 3, 4, 5, 6}
If the function call is `SumOfEven(arr, 2)`, the output should be 10 (as it sums 4 and 6).
If the function call is `SumOfEven(arr, 0)`, the output should be 12 (as it sums 2, 4, and 6).
If the function call is `SumOfEven(arr, 6)`, the output should be 0 (since the index is out of bounds). **[9 points]**

```
int SumOfEven(int[] arr, int index) {
      if (index >= arr.length)                                2 points
            return 0;
      if (arr[index] % 2 == 0)                                2 points
             return arr[index] + SumOfEven(arr, index + 1);   3.5 points
            else
                  return SumOfEven(arr, index + 1); }         1.5 points
```

---

2) Convert the following iterative (non-recursive) function into the equivalent recursive function. The function should count the odd numbers between 1 and n, where n is a positive integer.
[6 points]

<u>Iterative (non-recursive) function:</u>

```
public static int countOddNum(int n) {
         if (n < 1)
             return 0; // If n is less than 1, there are no odd numbers
         int count = 0;
         for (int i = 1; i <= n; i++)
             if (i % 2 != 0)
                 count++;
         return count;
     }
```

<u>Recursive function:</u>

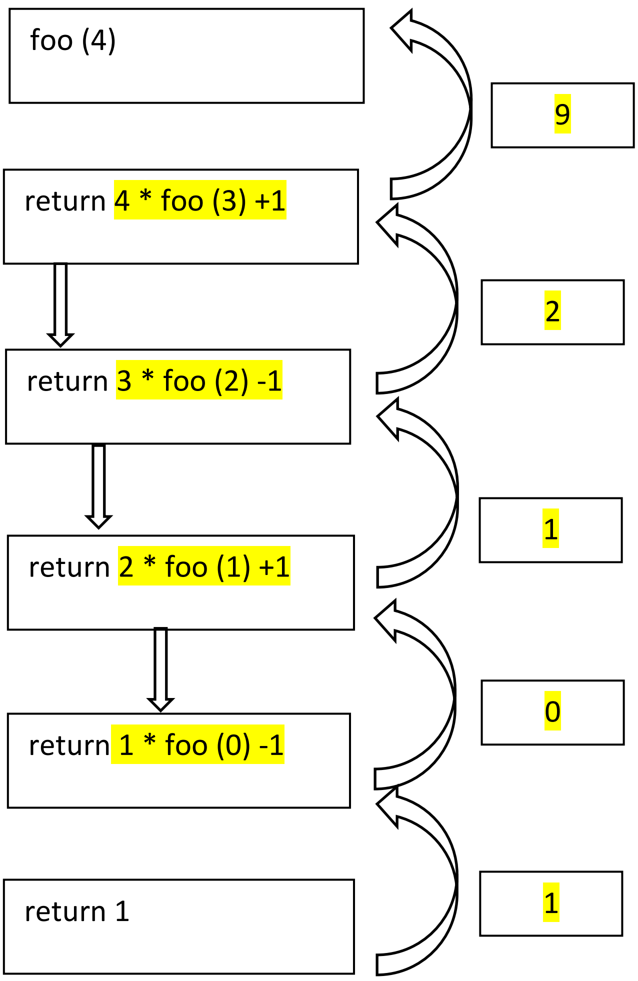
```
public static int countOddNum (int n) {
         if ( n < 1 )                            2 points
             return 0;
         if (n % 2 == 1)
             return  1 + countOddNum (n - 1)     2 points
            else
                  return countOddNum (n - 1); }  2 points
```

3) Consider the following recursive function `foo(int n)`. Determine the output printed by the call foo(4). **Show the final result, and function calls tracing in below diagram.** **[15 points]**

```
public class Main {
    public static void main(String[] args) {
        System.out.println("foo(4) = " + foo(4));
    }

    public static int foo(int n) {
        if (n == 0)
            return 1;
        else if (n % 2 == 1) {
            return n * foo(n - 1) - 1;
        else
            return n * foo(n - 1) + 1;
} }
```

2 points for each call → 8 points
1 point for each return → 5 points
2 points for final result → 2 points



foo (4) = 9

Question 4, Algorithm Analysis Topic: _________ / 12 points

1) Complete the following table that shows the running time against input size (n) of 3 algorithms and their running complexity expressed in O( ) notation. Then order the algorithms from fastest to slowest in terms of running complexity time. [6 points]

Algorithm	**n=2**	**n=3**	**n=4**	**Running Complexity**
A	12ms	12ms	12ms	O (1)
B	8ms	27ms	64ms	O (n^3)
C	4ms	8ms	16ms	O (2^n)

Fill in the blanks above for running complexities and complete the sentences below.

Fastest algorithm is algorithm **(A / B / C)**, the second fastest is algorithm **(A / B / C)**, and the slowest is algorithm **(A / B / C)**.

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2) An algorithm runs in O(log n) time. For input size of 32, the algorithm takes 10 milliseconds. How much time it will take for input size of 1024? [6 points]

T(n) = O(log n) = c* log(n)

T(32) = c*5 = 10ms → c = 2

T(1024) = c*10 = 20ms